

Unit tests 01 - Introduction

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... done with support of



Figure 1: scheidt&bachmann

Agenda

- ▶ Why tests are important
- ▶ Different types of testing
- ▶ Importance of unit testing
- ▶ How to write unit tests
- ▶ Best practices for unit testing

Why tests are important

- ▶ Risk Mitigation
 - ▶ Testing helps identify defects and failures early in the development process when they are less expensive to fix. This reduces project risks related to quality, security, performance, etc.
- ▶ Confidence
 - ▶ Executing a well-planned software test strategy provides confidence that the software works as intended before its release.
- ▶ Compliance
 - ▶ Testing can ensure that software adheres to standards, regulations, and compliance requirements. This is especially critical for safety-critical systems.
- ▶ User Satisfaction
 - ▶ Rigorous testing from a user perspective can verify usability, functionality, and compatibility. This increases customer/user satisfaction and reduces negative impacts to an organization's reputation or finances from poor quality products.

Why tests are important

- ▶ Optimization
 - ▶ Testing provides vital feedback that can be used to continuously improve software quality, user experience, security, performance and other product attributes.
- ▶ Cost Savings
 - ▶ Investing in testing activities reduces downstream costs related to defects found post-release. It is much cheaper to find and fix bugs earlier in the development cycle.

... and many more reasons

Why tests are important (SW developer perspective)

- ▶ Enable us testing corner case scenarios
 - ▶ Testing enables us to validate that our code behaves correctly under various conditions, including edge cases.
- ▶ Ensure code stability
 - ▶ Testing helps maintain code stability by catching regressions and unintended side effects early in the development process.
- ▶ Facilitate refactoring and code maintenance
 - ▶ Testing allows developers to confidently refactor and maintain code, knowing that existing functionality is protected by a suite of automated tests.

Why tests are important (SW developer perspective)

- ▶ They are instant!
 - ▶ Unit tests can be executed quickly (often in seconds or minutes), providing immediate feedback to developers about the correctness of their code.
- ▶ **Disclaimer**
 - ▶ They took time to write and maintain. Often it is 2x time of development. But AI can help significantly!
 - ▶ Unit tests are not a silver bullet; they complement other testing types and practices to ensure overall software quality.

Different types of (automated) testing

Unit tests Unit tests are very low level and close to the source of an application. They consist in testing individual methods and functions of the classes, components, or modules used by your software. Unit tests are generally quite cheap to automate and can run very quickly by a continuous integration server.

Integration tests Integration tests verify that different modules or services used by your application work well together. For example, it can be testing the interaction with the database or making sure that microservices work together as expected. These types of tests are more expensive to run as they require multiple parts of the application to be up and running.

Functional tests Functional tests focus on the business requirements of an application. They only verify the output of an action and do not check the intermediate states of the system when performing that action. There is sometimes a confusion between integration tests and functional tests as they both require multiple components to interact with each other. The difference is that an integration test may simply verify that you can query the database while a functional test would expect to get a specific value from the database as defined by